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TERA-print respectfully submits this comment to the Office of Science and Technology Policy in support of the development of America's AI Action Plan. The rapid innovation of AI seen thus far—and the need for America to maintain and extend its dominance and leadership in AI innovation and development—is of paramount importance to TERA-print. We are grateful that “it is the policy of the United States to sustain and enhance America's global AI dominance in order to promote human flourishing, economic competitiveness, and national security,”¹ and for the opportunity to contribute to the development of the AI Action Plan.

We are a Chicagoland startup spun out of the International Institute for Nanotechnology at Northwestern University, a globally recognized leader in nanotechnology, materials science, and chemistry. We design and build some of the world's most advanced technologies that enable massively parallel experimentation and fabrication at micro- and nanoscale – dramatically accelerating discovery and innovation across critical hard tech fields spanning biotechnology, advanced materials, and semiconductors, while generating unique massive datasets ideally suited for training next-generation domain-specific AI models.

TERA-print was founded in 2015 by Dr. Chad A. Mirkin, the George B. Rathmann Professor of Chemistry and Director of the International Institute for Nanotechnology at Northwestern University, and Dr. Andrey Ivankin, who led the development of the TERA-print's core ultrahigh-throughput experimentation technology. Our tools and technologies are already driving breakthroughs in high-impact research at leading universities, government laboratories, and top innovative startup across the United States and worldwide, and they are positioned to play an important role in strengthening America's leadership in the global hard tech AI race, ensuring long-term technological and economic dominance.

¹ *Removing Barriers to American Leadership in Artificial Intelligence*, Exec. Order No. 14,179, 90 Fed. Reg. 8,741 (Jan. 31, 2025).

In our comment, we outline the strategic importance of AI in advancing hard tech innovation, highlighting how this field differs from general-purpose AI due to the lack of large, diverse, and high-quality training datasets. We explain how TERA-print is developing technologies to bridge this gap, unlocking AI's full potential for materials discovery and disruptive technologies development. Finally, we provide key recommendations on the AI Action Plan, ensuring that America secures its leadership in this critical technological race.

Strategic Importance of AI for Hard Tech Innovation

Technological leadership in biotechnology, medical devices, microelectronics, and chemistry has long driven economic strength, national security, and global influence. From genomic datasets powering precision medicine to nanoscale materials enabling high-performance semiconductors, breakthroughs have consistently reshaped industries and fortified geopolitical standings.

Emerging hard tech innovations increasingly rely on miniaturization down to the micro- and nanoscale to unlock novel properties, enhance functionality, and significantly boost multiplexity and parallelization—enabling entirely new capabilities across industries. However, current micro- and nanofabrication methods remain costly, slow, and limited in scope – primarily optimized for semiconductor applications. Not only does this constrain the pace of hard tech innovation, but also leaves vast opportunities across biotechnology, medical devices, advanced materials, and beyond remain largely untapped.

AI has the potential to dramatically accelerate the design, optimization, and commercialization of breakthrough micro- and nanostructured technologies—unlocking novel properties, enhancing performance, and enabling entirely new capabilities across critical industries, all while significantly reducing reliance on costly and time-intensive physical experimentation.

Just as AI is enabling Apple and NVIDIA to design cutting-edge chips that dominate the industry, it has the potential to revolutionize biotechnology, pharmaceuticals, defense, and other hard tech sectors. For example, AI-driven design could help biotech and pharma companies develop more sensitive diagnostics and breakthrough therapies, while the military could leverage AI to create multifunctional devices for next-generation sensing, autonomous systems, and resilient battlefield technologies—enhancing national security and technological superiority.

Outside of the semiconductor industry and, to some extent, drug discovery, such advanced AI models either do not yet exist or remain severely limited in capability. However, the first to develop and deploy these models will gain a lasting competitive advantage—reshaping industries, strengthening national security, and defining the next era of technological dominance.

Data is the Bottleneck for the AI-Driven Hard Tech Industrial Revolution

AI models require extensive, high-quality datasets to generate accurate predictions. Yet, in most hard-tech sectors—including biotechnology, medical devices, microelectronics, and chemistry—data remain fragmented, scarce, and inconsistently structured, in stark contrast to the vast, readily available datasets that fuel general-purpose AI.

Additionally, as seen in advanced general-purpose AI models like ChatGPT, Gemini, and Claude, performance rapidly converges when trained on similar publicly available datasets—underscoring the strategic importance of proprietary data as the key differentiator. The same principle applies to domain-specific AI models—organizations that control the most comprehensive and proprietary datasets will gain an unrivaled competitive edge, drive future innovation, and solidify global technological leadership.

The U.S. must therefore close this data gap first to maintain its technological edge and prevent adversaries from gaining a strategic advantage.

The challenge, however, lies in the vast design universe of hard tech technologies—spanning materials and complex 3D architectures—which is extraordinarily diverse and nearly infinite. This makes it exceedingly difficult to generate adequate training datasets and develop transformative AI models that can generalize not even across, but within specific applications. Many of these problems are computationally intractable due to their hyperdimensionality. Miniaturization to the micro- and nanoscale in emerging hard tech innovations further complicates experimental prototyping and data generation, as traditional fabrication technologies are prohibitively slow, expensive, and restrictive—especially beyond semiconductor materials and applications.

This means that winning the AI—and, consequently, the broader technological—race in hard tech domains is not just about advancing algorithms or generating more data. The key lies in first developing generalizable physical-world technologies capable of producing training datasets with significantly higher throughput, versatility, and relevance to real-world applications.

TERA-print's Technologies for Unparalleled Large-Scale Datasets Generation and AI Advancement in Hard Tech: Powering America's Technological and Economic Dominance

TERA-print is pioneering the world's most advanced and versatile suite of tools, enabling massively parallel experimentation and prototyping with micro- to nano-precision across diverse materials and hard tech applications.

These tools empower researchers to conduct tens of thousands to millions of chemical syntheses experiments to develop novel materials or synthesize on demand DNA chips, prototype thousands of functional devices—such as sensors and optical components—or fabricate advanced multi-material structures with unprecedented properties, such as

those for tissue engineering or self-healing surfaces. What once took months or even years with current methods can now be achieved in just hours.

By integrating complementary high-throughput screening capabilities tailored to specific applications, these tools become the most powerful platforms—not only for experimental discovery and prototyping but also for the rapid generation of the largest training datasets in hard-tech domains where micro- and nanoscale precision is essential.

TERA-print's tools have already facilitated breakthroughs such as the discovery of novel drug delivery vehicles, brighter blue light LEDs, and new catalysts comprising lower critical materials for low intensity hydrogen production, just to name a few.

Remarkably, these tools cost only a fraction of that of advanced semiconductor fabrication systems, require no expensive or complex supporting infrastructure, and are compatible with a wide range of biological and organic materials—an inherent limitation in the semiconductor industry. Moreover, the underlying technology is highly scalable, enabling a 10- to 100-fold increase in the platform's data generation capabilities.

We strongly believe that massively parallel experimentation, fabrication, and characterization technologies—such as those developed by TERA-print and others—will be the cornerstone for generating the critical hard-tech datasets currently lacking. By enabling data creation at an unprecedented pace and scale, these technologies will unlock the most advanced AI models in this domain, securing the U.S.'s competitive dominance in the global technological race.

Recommendations for America's AI Action Plan

TERA-print believes that, among other important priorities, America's AI Action Plan should emphasize the following points.

Infrastructure. Many of the AI-based tools familiar to Americans, such as ChatGPT and consumer-facing platforms like Google Gemini, are trained on vast datasets sourced from publicly available internet content or open-domain repositories. However, applying AI to the cutting edge of technology demands specialized datasets that go beyond the public realm—tailored to highly specific and often sensitive conditions for a given application. To secure long-term leadership in AI-driven hard-tech innovation, investments must focus on the foundational systems, instruments, and technologies that enable the rapid generation of massive experimental datasets across a wide range of applications. By prioritizing physical-world technologies, the U.S. will safeguard its technological leadership from foreign imitators in ways that purely software-based solutions cannot.

Additionally, establishing Centers of Excellence—whether as standalone facilities, within national labs, or at universities—for large-scale data generation is essential to advancing the AI Action Plan in the context of disruptive hard technologies of the

future. Each Center would have a specialized focus—biotechnology, energy, aerospace, semiconductors—attracting the nation's most promising innovations to strengthen U.S. technological leadership and national security.

Moreover, there is a need to establish a standardized database architecture tailored to specific applications and domains, along with a centralized repository that enables multiple stakeholders to contribute and access high-quality datasets. This framework would facilitate collaboration, enhance interoperability, and accelerate AI-driven advancements across critical industries.

Investment. Just as physical infrastructure is essential for overcoming the challenges of data generation at scale, so too must the U.S. strengthen its investment apparatus to bridge the funding gap—commonly known as the “valley of death”—that exists between innovative startups and full-scale commercial and industrial deployment. The pace of development and the longer payoff timelines inherent to hard-tech breakthroughs often fall between the cracks of traditional venture capital models, which prioritize rapid returns, and existing grant mechanisms, which are often too incremental to support sustained, capital-intensive innovation. To maintain its technological edge, the U.S. must adapt its funding strategies by expanding long-term (patient) capital initiatives, leveraging the strengths of programs like the Small Business Innovation Research (SBIR) program, and directing strategic public investment toward long-term, high-impact AI and hard-tech innovations. A restructured investment framework that aligns with the realities of deep-tech development is critical to ensuring these transformative technologies reach market viability and reinforce national leadership.

Industrial Collaboration through Policy and Grants. Building an AI startup ecosystem capable of addressing industry's most pressing challenges requires a strategic approach to fostering collaboration between early-stage researchers in academia, government, and private industry. While necessity drives invention, innovators must have visibility into these critical needs—siloefforts risk slowing progress and undermining global competitiveness. The AI Action Plan should work to clearly communicate industry priorities and incentivize partnerships between industrial players and academic institutions, government agencies, small businesses, and other domestic entities. Additionally, the plan should integrate with existing funding mechanisms, prioritizing high-risk, high-reward projects—whether pursued independently or through industrial collaborations—to accelerate transformative breakthroughs.

These proposals aim to accelerate America's leadership in AI-driven hard-tech innovation by expanding massively parallel, real-world experimentation—unlocking unique, high-value datasets essential for training the most advanced AI models.

TERA-print has been a proud recipient of government small business grants through the National Science Foundation, the Department of Energy, and the Department of Defense over its lifetime. The work and experience gained through these programs has

exemplified the value of empowering small businesses and early-stage researchers to tackle pressing problems and develop powerful products to address industrial and government needs. The sum of these grants totals roughly \$2.9M over 10 years, but in the lifetime of these grants, TERA-print has generated multiple novel micro- and nanofabrication tools capable of generating massively parallel low-cost datasets and embarked on multiple collaborative projects with private and public entities that were manifested through the projects.²³ These projects are focused on high-impact areas such as biotech and semiconductor materials where accelerated materials discovery and device design are fueled by massive dataset generation, the exact type of projects that will unlock and be revolutionized by AI.

Expanding federal investment in large-scale data generation technologies for hard tech is essential to maintaining America's leadership in AI and advanced technology, reducing its reliance on foreign technologies, and accelerating the development of disruptive technologies of the future critical to defense, biotechnology, and industry.

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In summary, TERA-print is a tremendous believer in America's efforts and abilities to spur an industrial revolution with AI and establish a new global dominance in doing so. We are ready to support these efforts and help accomplish America's goals in relation to the AI Action Plan.

Very truly yours,

Dr. Andrey Ivankin
Co-Founder
TERA-print LLC

² TERA-print LLC | SBIR. (n.d.). <https://www.sbir.gov/portfolio/860753>

³ U.S. Department of Defense. (n.d.). DOD announces five DESI Awards for University-Industry Collaborations. <https://www.defense.gov/News/Releases/Release/Article/1594969/dod-announces-five-desi-awards-for-university-industry-collaborations/>